

浙江财经大学

课后作业

作业名称 Practices of Relational Algebra
and Tuple Relational Calculus

所属课程名称 Database Theory and Application

学 院 Information Technology and Artificial Intelligence

班 级 _____

学 号 _____

姓 名 _____

指导教师 Deng Shengchun

CAP Database

CUSTOMERS	A table containing information about customers
cid	Unique identifier for a customer/row – Note there is no customer corresponding to cid='C005'
cname	Name of a customer
city	City where the customer (headquarters) is located
discent	Each customer has a negotiated discount on prices
AGENTS	A table containing information for agent
aid	Unique identifier for an agent/row
aname	Lname of an agent
city	City where the agent is based
percent	Percentage commission each agent receives on each sale
PRODUCTS	A table containing information about products for sale
pid	Unique identifier for a product/row
pname	Name of a product
city	City where this product is warehoused
quantity	Quantity on hand for sale, in standard units
price	Wholesale price of each unit product
Note that the same column name, city, appears in all three tables defined so far. This is not a coincidence.	
ORDERS	A table containing information about orders
ordno	Unique identifier for this order
month	Month the order was placed; assume that orders started in January of this year
cid	This customer...
aid	...purchased through this agent...
pid	...this specific product...
qty	...in this total quantity...
dollars	...at this dollars cost

CUSTOMERS

cid	cname	city	discent
c001	TipTop	Duluth	10.00
c002	Basics	Dallas	12.00
c003	Allied	Dallas	8.00
c004	ACME	Duluth	8.00
c006	ACME	Kyoto	0.00

AGENTS

aid	aname	city	percent
a01	Smith	New York	6
a02	Jones	Newark	6
a03	Brown	Tokyo	7
a04	Gray	New York	6

a05	Otasi	Duluth	5
a06	Smith	Dallas	5

PRODUCTS

pid	pname	city	quantity	price
p01	comb	Dallas	111400	0.50
p02	brush	Newark	203000	0.50
p03	razor	Duluth	150600	1.00
p04	pen	Duluth	125300	1.00
p05	pencil	Dallas	221400	1.00
p06	folder	Dallas	123100	2.00
p07	case	Newark	100500	1.00

ORDERS

ordno	month	cid	aid	pid	qty	dollars
1011	Jan	c001	a01	p01	1000	450.00
1012	Jan	c001	a01	p01	1000	450.00
1019	Feb	c001	a02	p02	400	180.00
1017	Feb	c001	a06	p03	600	540.00
1018	Feb	c001	a03	p04	600	540.00
1023	Mar	c001	a04	p05	500	450.00
1022	Mar	c001	a05	p06	400	720.00
1025	Apr	c001	a05	p07	800	720.00
1013	Jan	c002	a03	p03	1000	880.00
1026	May	c002	a05	p03	800	704.00
1015	Jan	c003	a03	p05	1200	1104.00
1014	Jan	c003	a03	p05	1200	1104.00
1021	Feb	c004	a06	p01	1000	460.00
1016	Jan	c006	a01	p01	1000	500.00
1020	Feb	c006	a03	p07	600	600.00
1024	Feb	c006	a06	p01	800	400.00

Solve in relational algebra and tuple relational calculus for the following queries involving the CAP database.

(1) Find all (ordno, pid) pairs for orders of quantity equal to 1000 or more.

(1) 查询订单中物品数量大于或者等于 1000 的(ordno, pid)对

$\pi_{\text{ordno, pid}}(\sigma_{\text{quantity} \geq 1000}(\text{orders}))$

$\{ o[\text{ordno, pid}] \mid o \in \text{orders} \wedge o[\text{quantity}] \geq 1000 \}$

(2) Find all product names of products priced between \$0.50 and \$1.00, inclusive.

(2) 查询所有价格在\$0.50 和\$1.00 之间的商品名字，包括边界价格

$\pi_{pname}(\sigma_{price \geq 0.50 \wedge price \leq 1.00}(products))$

$\{ p[pname] \mid p \in Products \wedge (p[price] \geq 0.5 \wedge p[price] \leq 1.0) \}$

(3) Find all (ordno, cname) pairs for orders of dollar value less than \$500.

(3) 查询订单价格低于\$500 的(ordno, cname)对

$\pi_{ordno, cname}(\sigma_{dollars < 500}(Customers \bowtie Orders))$

$\{ o[ordno], c[cname] \mid (o \in Orders \wedge o[dollars] < 500) \wedge (c \in Customers \wedge o[cid] = c[cid]) \}$

(4) Find all (ordno, aname) pairs for orders in March.

(4) 查询所有三月份接受的订单的(ordno, aname)对

$\pi_{ordno, aname}(\sigma_{month='mar'}(Agents \bowtie Orders))$

$\{ o[ordno], a[aname] \mid (o \in Orders \wedge o[month] = 'mar') \wedge (a \in Agents \wedge o[aid] = a[aid]) \}$

(5) Find all (ordno, cname, aname) triples for orders in March.

(5) 查询所有三月份接受的订单的(ordno, cname, aname)三元组

$\pi_{ordno, cname, aname}(\sigma_{month='mar'}(Customers \bowtie Orders) \bowtie \pi_{aid, aname}(Agents))$

$\{ o[ordno], c[cname], a[aname] \mid o \in Orders \wedge c \in Customers \wedge a \in Agents \wedge o[cid] = c[cid] \wedge o[aid] = a[aid] \wedge o[month] = 'mar' \}$

(6) Find all the names of agents in New York who placed orders of individual dollar value less than \$500.

(6) 查询所有位于 New York 的代理商，并且要求这些代理商所接受的单个订单价格少于 \$500 [只要有一个订单满足即可]

$\pi_{aid, aname, city, percent}(\sigma_{city='New York' \wedge dollars < 500}(Agents \bowtie Orders))$

$\{ a \mid a \in Agents \wedge a[city]='New York' \wedge o \in Orders \wedge a[aid] = o[aid] \wedge o[dollars] < 500 \}$

(7) Find all product names of products in Duluth ordered in March.

(7) 查询所有三月份定购的(位于)Duluth 的商品的名字

$\pi_{pname}(\sigma_{month='mar' \wedge city='Dallas'}(Products \bowtie Orders))$

$\{ p[pname] \mid p \in Products \wedge p[city]='Dallas' \wedge o \in Orders \wedge o[pid] = p[pid] \wedge o[month] = 'mar' \}$

(8) Find all (cid, aid, pid) triples for customer, agent, product combinations that are all in the same city.

(8) 查询所有顾客、代理商和商品都在同一个城市的三元组(cid, aid, pid)

$\pi_{cid, aid, pid} (Customers \bowtie Agents \bowtie Products)$

$\{ c[cid], a[aid], p[pid] \mid c \in Customers \wedge a \in Agents \wedge p \in Products$
 $\wedge c[city] = a[city] \wedge c[city] = p[city] \}$

(9) Find all (cid, aid, pid) triples for customer, agent, product combinations that are not all in the same city (any two maybe).

(9) 查询所有顾客、代理商和商品不都在同一个城市(可能有两个在同一城市)的三元组(cid, aid, pid)

$\pi_{cid, aid, pid} (Customers \times Agents \times Products)$
 $- \pi_{cid, aid, pid} (Customers \bowtie Agents \bowtie Products)$

$\{ c[cid], a[aid], p[pid] \mid c \in Customers \wedge a \in Agents \wedge p \in Products$
 $\wedge \neg (c[city] = a[city] \wedge c[city] = p[city] \wedge a[city] = p[city]) \}$

(10) Find all (cid, aid, pid) triples for customer, agent, product combination, no two of which are in the same city.

(10) 查询所有顾客、代理商和商品两两不在同一个城市的三元组(cid, aid, pid)

$\pi_{cid, aid, pid} (\sigma_{Customers.city \neq Agents.city \wedge Customers.city \neq Products.city \wedge Products.city \neq Agents.city} (Customers \times Agents \times Products))$

$\{ c[cid], a[aid], p[pid] \mid c \in Customers \wedge a \in Agents \wedge p \in Products$
 $\wedge (c[city] \neq a[city] \wedge c[city] \neq p[city] \wedge a[city] \neq p[city]) \}$

(11) Get cities of agents booking an order from customer c002.

(11) 查询接受顾客 c002 订单的代理商所在的城市

$\pi_{city} (\pi_{aid} (\sigma_{cid='c002'} (Orders)) \bowtie Agents)$

$\{ a[city] \mid a \in Agents \wedge o \in Orders \wedge (o[aid] = a[aid] \wedge o[cid] = 'c002') \}$

(12) Get product names ordered by at least one customer based in Dallas through an agent based in Tokyo.

(12) 查询至少被一个在 Dallas 的顾客通过位于 Tokyo 的代理商定购的商品的名字

$$\pi_{pname}(((\pi_{aid, pid}((\pi_{cid}(\sigma_{city='Dallas'}(Customers)) \bowtie Orders)) \bowtie (\pi_{aid}(\sigma_{city='Tokyo'}(Agents)) \bowtie Products)))$$

$$\{ p[pname] \mid p \in Products \wedge \exists (c \in Customers \wedge c[city]='Dallas') \wedge (a \in Agents \wedge a[city]='Tokyo') \wedge (o \in Orders))$$

$$(o[aid]=a[aid] \wedge o[cid]=c[cid] \wedge o[pid]=p[pid]) \}$$

(13) Get **pids** of products ordered through any agent who makes at least one order for a customer in Dallas.

(13) 查询曾经收到 Dallas 的顾客订单的代理商所销售的所有的商品的 pid 值

$$\pi_{pid}(Orders \bowtie (\pi_{aid}(\sigma_{city='Dallas'}(Customers)) \bowtie Orders))$$

$$\{ p[pid] \mid p \in Products \wedge \exists (c \in Customers \wedge c[city]='Dallas') \wedge (\exists (o \in Orders) (o[pid]=p[pid] \wedge o[cid]=c[cid])) \}$$

(14) Display all pairs of **aids** for agents who live in the same city.

(14) 查询所有在同一个城市的代理商的 aid 对

$$\pi_{a1.aid, a2.aid}(\sigma_{a1.city=a2.city \wedge a1.aid > a2.aid}(\rho_{a1}(Agents) \times \rho_{a2}(Agents)))$$

$$\{ a1[aid], a2[aid] \mid a1 \in Agents \wedge a2 \in Agents \wedge a1[city] = a2[city] \wedge a1[aid] > a2[aid] \}$$

(15) Find **cids** of customers who did not place an order through agent a03.

(15) 查询没有通过代理商 a03 订购过商品的顾客的 cid 值

$$\pi_{cid}(Customers) - \pi_{cid}(\sigma_{aid='a03'}(Orders))$$

$$\{ c[cid] \mid c \in Customers \wedge \neg \exists (o \in Orders \wedge o[cid]=c[cid]) (o[aid] = 'a03') \}$$

(16) Find **cids** of customers who have the largest discount; separately, find those who have the smallest discount.

(16) 查询折扣率最大和最小的顾客的 cid 值

$$\pi_{cid}(Customers) - \pi_{c1.cid}(\sigma_{c1.discnt < c2.discnt}(\rho_{c1}(Customers) \times \rho_{c2}(Customers)))$$

$$\pi_{c1.cid, c2.cid}(\sigma_{c1.discnt >= c2.discnt}(\rho_{c1}(Customers) \times \rho_{c2}(Customers)))$$

$$\div \pi_{cid}(Customers)$$

$$\{ c[cid] \mid c \in Customers \wedge \neg \exists (c1 \in Customers) (c1[discnt] > c[discnt]) \}$$

(17) Find **cids** of customers who order all products.

(17) 查询订购了所有商品的顾客的 cid 值

$$\pi_{cid, pid}(\text{Orders}) \div \pi_{pid}(\text{Products})$$

$$\{c[cid] \mid c \in \text{Customers} \wedge \forall (p \in \text{Products})$$

$$(\exists(o \in \text{Orders})(o[pid]=p[pid] \wedge o[cid]=c[cid]))\}$$

(18) Find **pids** of products ordered through agent a03 but not through agent a06.

(18) 查询通过代理商 a03 而不通过代理商 a06 订购的商品的 pid 值

$$\pi_{pid}(\sigma_{aid='a03'}(\text{Orders})) - \pi_{pid}(\sigma_{aid='a06'}(\text{Orders}))$$

$$\{o[pid] \mid o \in \text{Orders} \wedge o[aid] = 'a03' \wedge \neg \exists(o1 \in \text{Orders} \wedge o1[pid]=o[pid]) (o1[aid] = 'a06')\}$$

(19) Get **pnames** and **pids** of products that are stored in the same **city** as one of the agents who sold these products.

(19) 查询商品的 pname 和 pid 值，要求这些商品所在的城市和某个销售过该商品的代理商所在的城市相同

$$\pi_{pname, pid}(\text{Products} \bowtie \pi_{pid, city}(\text{Orders} \bowtie \text{Agents}))$$

$$\{p[pname, pid] \mid p \in \text{Products} \wedge (a \in \text{Agents} \wedge a[city]=p[city]) \wedge \exists(o \in \text{Orders}) (o[pid] = p[pid] \wedge o[aid] = a[aid])\}$$

(20) Get **aids** and **anames** of agents with **aname** beginning with the letter “S” who do not place orders for any product in Duluth.

(20) 查询名字是以 S 开头的代理商的 aid 和 aname 值，并且这些代理没有销售过任何 Duluth 生产的商品

$$\pi_{aid, aname}(\sigma_{aname \geq 'S' \wedge aname < 'T'} \text{Agent} \bowtie (\pi_{aid}(\text{Agents}) - \pi_{aid}(\sigma_{city='Duluth'}(\text{Products} \bowtie \text{Orders}))))$$

$$\{a[aid, aname] \mid a \in \text{Agents} \wedge a[aname] \geq 'S' \wedge a[aname] < 'T' \wedge \forall(p \in \text{Products} \wedge p[city]='Duluth') (\neg \exists(o \in \text{Orders}) (o[aid] = a[aid] \wedge o[pid] = p[pid]))\}$$

(21) Get **cids** of customers who order both product p01 and product p07.

(21) 查询同时订购了商品 p01 和 p07 的顾客的 cid 值

$$\pi_{o1.cid}(\sigma_{o1.cid=o2.cid \wedge o1.pid='p01' \wedge o2.pid='p07'}(\rho_{o1}(\text{Orders}) \times \rho_{o2}(\text{Orders})))$$

$$\{o[cid] \mid o \in \text{Orders} \wedge o[pid] = 'p01' \wedge \exists(o1 \in \text{Orders}) (o1[cid] = o[cid] \wedge o1[pid] = 'p07')\}$$

(22) Get names of agents who place orders for all products ordered by customer c002.

(22) 查询销售过所有曾被顾客 c002 订购过的商品的代理商的名字

$$\pi_{aname} (Agents \bowtie (\pi_{aid, pid} (Orders) \div \pi_{pid} (\sigma_{cid='c002'} (Orders))))$$

$$\{a[aname] \mid a \in Agents \wedge \forall (o1 \in Orders \wedge o1[cid]='c002') \\ (\exists (o2 \in Orders) (o2[aid] = a[aid] \wedge o1[pid] = o2[pid]))\}$$

- (23) Get names of agents who place orders for all products that are ordered by any customer at all. (Hint: The phrase “any customer at all” means the same as “some customer.”)

- (23) 查询销售过所有曾被某些顾客订购过的商品的代理商的名字

$$\pi_{aname} (Agents \bowtie (\pi_{aid, pid} (Orders) \div \pi_{pid} (Orders)))$$

$$\{a[aname] \mid a \in Agents \wedge \forall (o1 \in Orders) \\ (\exists (o2 \in Orders) (o2[aid] = a[aid] \wedge o1[pid] = o2[pid]))\}$$

- (24) Get (cid, aid, pid) triples for customer, agent, product combination so that at most two of them are in the same city.

- (24) 查询所有的三元组(cid, aid, pid)，要求对应的顾客，代理商和商品中最多有两者是位于同一座城市

$$\pi_{cid, aid, pid} (Customers \times Agents \times Products) -$$

$$\pi_{cid, aid, pid} (\sigma_{Customers.city \neq Agents.city \wedge Customers.city \neq Products.city \wedge \\ Products.city \neq Agents.city} (Customers \times Agents \times Products))$$

$$\{c[cid], a[aid], p[pid] \mid c \in Customers \wedge a \in Agents \wedge p \in Products\} \wedge \\ \neg (c[city] \neq a[city] \wedge c[city] \neq p[city] \wedge a[city] \neq p[city])\}$$

- (25) Get pids of products ordered by all customers who place any order through agent a03.

- (25) 查询所有曾在代理商 a03 处订购商品的顾客订购过的商品的 pid 值

$$\pi_{pid} (Orders \bowtie (\pi_{cid} (\sigma_{aid='a03'} (Orders))))$$

$$\{o1[pid] \mid o1 \in Orders \wedge \exists (o2 \in Orders \wedge o2[aid] = 'a03') (o2[cid] = o1[cid])\}$$

- (26) Get aids of agents who place individual orders in dollar value greater than \$500 for customer living in Dallas.

- (26) 查询接受过 Dallas 的顾客一笔总额超过\$500 的订单的代理商的 aid 值

$$\pi_{aid} (\sigma_{city='Dallas' \wedge dollars > 500} (Customer \bowtie Orders))$$

$$\{o[aid] \mid o \in Orders \wedge o[dollars] > 500 \wedge$$

$$\exists (c \in Customers \wedge c[city] = 'Dallas') (c[cid] = o[cid])\}$$

- (27) Give all (cname, aname) pairs where the customer places an order through the agent.

- (27) 查询所有的(cname, aname)对，要求对应的顾客曾经在对应的代理商处订购过商品

$$\pi_{cname, aname}(\pi_{cid, cname}(\text{Customers}) \bowtie \text{Orders} \bowtie \pi_{aid, aname}(\text{Agents}))$$

$$\{c[cname], a[aname] \mid c \in \text{Customers} \wedge o \in \text{Orders} \wedge a \in \text{Agents} \wedge (o[cid] = c[cid] \wedge o[aid] = a[aid])\}$$

(28) Get **cids** of customers who order all their products through only one agent.

(28) 查询只从一家代理商订购过商品的顾客的 cid 值

$$\pi_{cid}(\text{Orders}) - \pi_{o1.cid}(\sigma_{o1.cid=o2.cid \wedge o1.aid \neq o2.aid}(\rho_{o1}(\text{Orders}) \times \rho_{o2}(\text{Orders})))$$

$$\{o1[cid] \mid o1 \in \text{Orders} \wedge \neg \exists (o2 \in \text{Orders}) (o1[cid] = o2[cid] \wedge o1[aid] \neq o2[aid])\}$$

(29) Get **aids** of agents who do not provide any products produced in Dallas for any customer living in Duluth.

(29) 查询没有为任何住在 Duluth 的顾客订购任何在 Dallas 生产的产品的代理商 aid 值

$$\pi_{aid}(\text{Agents}) - \pi_{aid}(\sigma_{\text{Customers.city}='Duluth' \wedge \text{Products.city}='Dallas'}(\text{Customers} \times \text{Products} \times \text{Orders}))$$

$$\{a[aid] \mid a \in \text{Agents} \wedge \forall (c \in \text{Customers} \wedge c[city]='Duluth') \wedge \forall (p \in \text{Products} \wedge p[city]='Dallas') (\neg \exists (o \in \text{Orders}) (o[aid] = a[aid] \wedge o[cid] = c[cid] \wedge o[pid] = p[pid]))\}$$

(30) Get **aids** of agents who provide at least one same product for all customers living in Duluth or customers living in Kyoto.

(30) 查询为住在 Duluth 或 Kyoto 的所有顾客订购了至少一种同样产品的代理商的 aid 值

$$\pi_{aid}((\pi_{aid, pid}(\text{Orders}) \div \pi_{pid}(\sigma_{city='duluth'}(\text{Customers})) \bowtie \text{Orders})) \cup \pi_{aid}((\pi_{aid, pid}(\text{Orders}) \div \pi_{pid}(\sigma_{city='kyoto'}(\text{Customers})) \bowtie \text{Orders}))$$

$$\{o[aid] \mid (o \in \text{Orders} \wedge \forall (c \in \text{Customers} \wedge c[city]='Duluth') (\exists (o1 \in \text{Orders}) (o1[cid] = c[cid] \wedge o1[pid] = o[pid]))) \vee (o \in \text{Orders} \wedge \forall (c \in \text{Customers} \wedge c[city]='Kyoto') (\exists (o1 \in \text{Orders}) (o1[cid] = c[cid] \wedge o1[pid] = o[pid])))\}$$

(31) Get **cids** of customers who order product only from agent a03 or a05.

(31) 查询只通过代理商 a03 或 a05 订货的顾客的 cid 值

$$\pi_{cid}(\sigma_{aid='a03' \vee aid='a05'}(\text{Orders})) - \pi_{cid}(\sigma_{aid \neq 'a03' \wedge aid \neq 'a05'}(\text{Orders}))$$

$$\pi_{cid}(\sigma_{aid='a03' \vee aid='a05'}(\text{Orders})) - \pi_{o1.cid}(\sigma_{o1.aid \neq 'a03' \wedge o2.aid \neq 'a05' \wedge o2.cid=o1.cid}(\rho_{o1}(\text{Orders}) \times \rho_{o2}(\text{Orders})))$$

$$\{c[cid] \mid c \in \text{Customers} \wedge \forall (o1 \in \text{Orders} \wedge o1[cid]=c[cid] \wedge (o1[aid]='a03' \vee o1[aid]='a05')) (\neg \exists (o2 \in \text{Orders} \wedge o2[cid]=c[cid]) (o2[aid] \neq 'a03' \vee o2[aid] \neq 'a05'))\}$$

(32) Get **pids** of products ordered by all customers living in Duluth.

(32) 查询被所有住在 Duluth 的顾客都订购了的产品 pid 值

$$\pi_{pid, cid}(\mathbf{Orders}) \div \pi_{cid}(\sigma_{city='Duluth'}(\mathbf{Customers}))$$

$$\{ p[pid] \mid p \in \mathbf{Products} \wedge \forall (c \in \mathbf{Customers} \wedge c[city]='Duluth') \\ (\exists (o \in \mathbf{Orders})(o[pid]=p[pid] \wedge o[cid]=c[cid])) \}$$

(33) Get aids of agents who have the highest Percent.

(33) 查询拥有最高 Percent 的代理商

$$\pi_{aid}(\mathbf{Agents}) - \pi_{a1.aid}(\sigma_{a1.percent < a2.percent} (\rho_{a2}(\mathbf{Agents}) \times \rho_{a2}(\mathbf{Agents})))$$

$$\{ a1[aid] \mid a1 \in \mathbf{Agents} \wedge \neg \exists (a2 \in \mathbf{Agents})(a2[percent] > a1[percent]) \}$$